EE1003
Introduction to Computer I

PA2: Hint

Andy, Yu-Guang Chen
Associate Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw
Slides Credit: TA Tse-Chuan Yang



Remind

This assignment requires you to complete a game based on the Knight's Tour problem.

- **PA2 Due Date:** November 19
- **Readability Requirement:** Add brief English comments to explain your code clearly.
- **Report Focus:** Emphasize how your **recursion** and **pruning strategy** are designed and implemented.



Hint

Basic:

- You can use a **N*N 2D array** to represent the **N*N** board.
- You can define the **knight's 8 moves** in a const array.

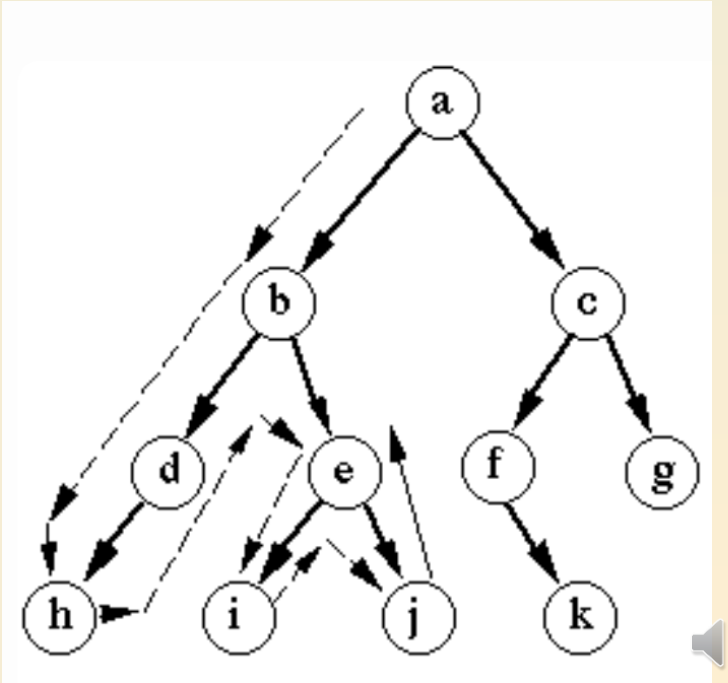
Advanced:

- To make your code cleaner, use **pair<int,int>** to represent coordinates.
- For flexibility, you may use **vector** (dynamic array) instead of fixed-size array.



Hint

- Use **recursion (DFS)** to explore all possible routes that the knight can take.
- Each recursive call represents one move to a new square.
- When there are no valid moves left, backtrack to explore other possibilities.



Hint

- Too slow? You need **pruning** to cut off unpromising paths early.
- Always **update your best score** whenever you find a better solution.
- Stop recursion **immediately** if *(current score + remaining maximum potential < best score).*







Hint

Advanced:

- Furthermore, you can decide **which next move to explore first** in recursion by defining your own **cost function**, and sort the moves accordingly. (Insertion sort is simple and efficient here.)
- You may apply **Warnsdorff's rule** to speed up the search in a certain **hidden test case**.





Bonus Hint

- Andy's blocking zone **expands over time**, so plan your route carefully, stop recursion **if the next move will soon be blocked** by Andy's expansion.
- You need to apply **pruning carefully.** Andy's blocking zone expands over time, which will gradually reduce your available search space and **shrink the upper bound**.







Notice

- Hidden cases may be **much harder**, so make sure your method is **efficient and well-designed**.
- We will inspect your code**,** don't **over-prune** even if your answer seems correct.





Q&A

- If you encounter any issues, feel free to email me for clarification or assistance.
- TA: Tse-Chuan Yang
- TA email is kevin200481211@gmail.com



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Thank you!





